MILITARY SPECIFICATION

FUZE, PIBD, M550; TEST VEHICLE FOR METAL PARTS AND LOADING, ASSEMBLING AND PACKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not normally covered by the drawings and provides quality assurance provisions for the fabrication of parts and the loading, assembling and packing for one type of test vehicle for use with the Fuze, PIBD, M550.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078	– Ammunition, Standard Quality Assurance Provision, General Specification for

DRAWINGS

U.S. ARMY ARMAMENT COMMAND

PRODUCT AND PACKAGE DRAWINGS

9287616	– Test Vehicle for M550 Fuze
9288856	– Box, Fiberboard, Inner Pack, Ammunition for Test Vehicle for Fuze M550
9288857	– Box, Fiberboard, Outer Pack, Ammunition for Test Vehicle for Fuze M550

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1390

AMSC N/A

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INSPECTION EQUIPMENT DRAWINGS

8827889 - Chamber Gage

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or requests for proposal shall apply.

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 0-190

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify, "49 CFR 0-190 (latest revision)")

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts.-The parts shall comply with all requirements specified on Drawing (Dwg) 9287616, all associated drawings, and with all requirements specified in applicable specifications.

3.3 Workmanship.-All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. exterior surface coatings shall be continuous; however, a few light scratches not exposing base material may be permitted. All required marking and stamping shall be neat and sharply defined.

3.4 Submission of results of contractor-conducted examinations and tests.-The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract. (See 6.4)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-The responsibility for inspection shall comply with MIL-A-48078.

4.2 Classification of Inspection.-The following type of inspection shall be conducted on this item:

a. Quality Conformance Inspection

4.3 First Article Inspection.-Not applicable

4.4 Quality Conformance Inspection

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4.4.1 Inspection Lot Formation.-See MIL-A-48078. Each inspection lot shall contain:

a. Cartridge case loaded assemblies from one interfix lot number from one manufacturer.

b. Projectile metal parts from one interfix lot number from one manufacturer.

c. Primers from one lot number from one manufacturer.

d. Propellant from not more than one lot number from one manufacturer.

4.4.2 Examination.-See MIL-A-48078.

Sampling Plans.-Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE			DRAWING NUMBER
4.4.2.1	Test Vehicle		SHEET 1 OF 1	9287616 NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
101.	Depth to charge surface		0.40%	3.2	Gage
102.	Concentricity of charge with small inside diameter of skirt		0.40%	3.2	Gage
103.	Diameter of rotating bands		0.40%	3.2	Gage
104.	Security of test vehicle	80/ Lot	1.5%	3.2	4.5.4
105.	Chamber gage		100%	3.2	4.5.6
106.	Air pressure	200/ lot	1.5%	3.2	4.5.7
Minor					
201.	Total length		0.65%	3.2	Gage
202.	Perpendicularity of charge with small inside dia of skirt		0.65%	3.2	Gage
203.	Depth to bottom of cavity		0.65%	3.2	Gage
204.	Marking misleading or unidentifiable		0.65%	3.2	Visual
205.	Evidence of poor workmanship		0.65%	3.3	Visual
206.	Color of skirt incorrect		0.65%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9288856
4.4.2.2	Unsealed Fiber Box (Inner)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major					
101.	Test vehicle inverted		0.40%		Visual
102.	Incorrect number of test vehicles		0.40%		Visual
103.	Fillers missing		0.40%		Visual
Minor:	None defined				
5					

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE				DRAWING NUMBER
4.4.2.3	Sealed Fiber Box (Outer)		SHEET 1 OF 1		9288857 NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	Container damaged		0.40%	--	Visual
102.	Corners not blunted		0.40%	--	Visual
Minor					
201.	Contents loose		0.65%	--	Manual
202.	Label missing or unidentifiable		0.65%	--	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74

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4.4.3 Testing.-

4.4.3.1 Specific gravity of inert material (See Dwg. 9287617), Major defect.-Two (2) sample cup assemblies shall be selected from each four (4) hour's production and subjected to this test. If any sample fails to comply with the specific gravity requirement specified on the applicable drawing, the four (4) hour's production represented by the sample shall be rejected.

4.4.4 Inspection Equipment.-The inspection equipment require to perform the examination and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with Paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078((PA).

4.5 Test Methods and Procedures.-

4.5.1 Specific gravity of inert material.-The specific gravity of the inert material shall be determined as follows:

Weigh the empty cup and skirt assembly in grams and record as We. Fill the empty assembly with water at 75 degrees ± 2 degrees F., weigh in grams and record the weight as Ww. Determine the volume by substracting We from Ww. Load the assembly with inert material, drill the Liner assembly cavity, weigh in grams and record the weight as W1. Determine the inert material weight by subtracting We from W1. Calculate the specific gravity as follows:

$$\text{Specific gravity} = \frac{W1 - We}{Ww - We}$$

4.5.2 Security of test vehicle.-The test vehicle shall be placed in an approved fixture and the force specified on Dwg. 9287616 shall be applied. Cartridge cases shall not be reused. Test vehicles may be reused.

4.5.3 Chamber gage failure.-The test vehicle shall be placed in the chamber gage (Dwg 8827889) and must meet the requirements specified on Dwg. 9287616. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.5.4 Air pressure test of test vehicle.-The test vehicle shall be placed in an approved fixture and a measured quantity of air shall be applied to produce the required air pressure. This test is a non-destructive test. Parts so tested may be returned to the lot.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 Level C.-The test vehicle shall be preserved and packaged in accordance with Dwg. 9288856.

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5.2 Packing.-

5.2.1 Level C.-The test vehicle, preserved and packaged as specified in 5.1.1, shall be packed for shipment in accordance with Dwg. 9288857.

5.3 Marking.-Marking shall be in accordance with Dwgs. 9288856, 9288857 and Code of Federal Regulations, Title 49, Parts 0-190.

5.4 Shipping.-When components from more than one lot are shipped as a carload, each lot shall be kept separate and the division between lots clearly indicted to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use.-The components covered by this specification are intended for use with the Fuze, PIBD, M550.

6.2 Ordering data.-See MIL-A-48078.

6.3 Submission of Designs for Approval.-See 6.2.3 of MIL-A-48078. Submit equipment designs, as required, to Commander, Picatinny Arsenal, Attn: SARPA-QA-A-S, Dover, New Jersey 07801.

6.4 Submission of results of contractor conducted examinations and tests.-Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, Attn: SARPA-QA-A-S.

Custodian:
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Preparing Activity:
Army-AR

Project Number: 1390-A029

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	Mil-F-48275 (AR)	740318

3. DOCUMENT TITLE: FUZE, PIBD, M550; TEST VEHICLE FOR METAL PARTS AND LAP

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial 201-724-6675	(2) AUTOVON DSN-880-6675
c. ADDRESS (Include Zip Code) ATTN: SMACA-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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